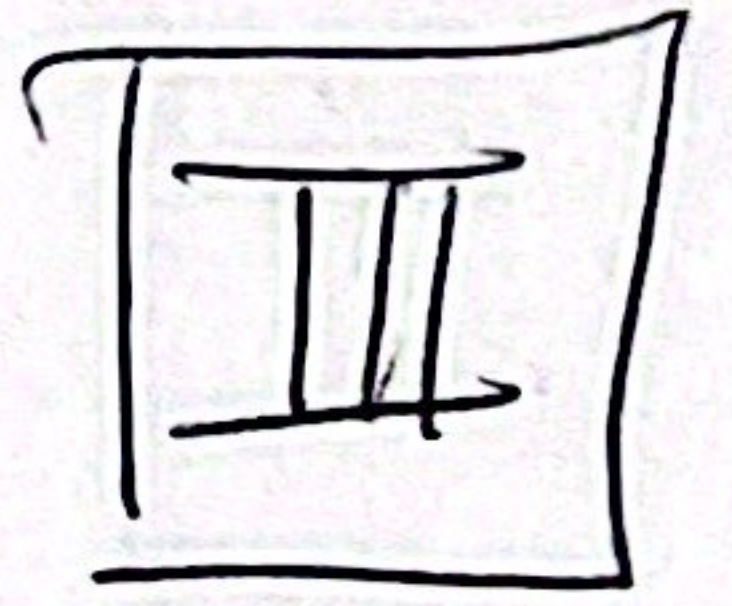




Kinetics — Linear motion.

The concept of speed is familiar to everyone. It is the rate at which something moves. (or rate of change of position with time)

$$\text{Average speed} = \frac{\text{total distance traveled } (x_1 - x_0)}{\text{total elapsed time } (t_1 - t_0)}$$

$$\bar{v} = \frac{\Delta x}{\Delta t} \quad \text{units } \underline{\underline{m/s}}$$

In general, for any small interval of distance Δx that is traversed in the time interval Δt the average speed is given by

$$\bar{v} = \frac{\Delta x}{\Delta t}$$

Example.

In a two part, 30 miles trip; 15 miles at 60 miles/hr and 15 miles at 20 miles/hr at total time 1 hr.

The average speed is $\bar{v} = \frac{x}{t} = \frac{x_1 + x_2}{t_1 + t_2} = \frac{15 \text{ mi} + 15 \text{ mi}}{\frac{1}{4} \text{ hr} + \frac{3}{4} \text{ hr}}$

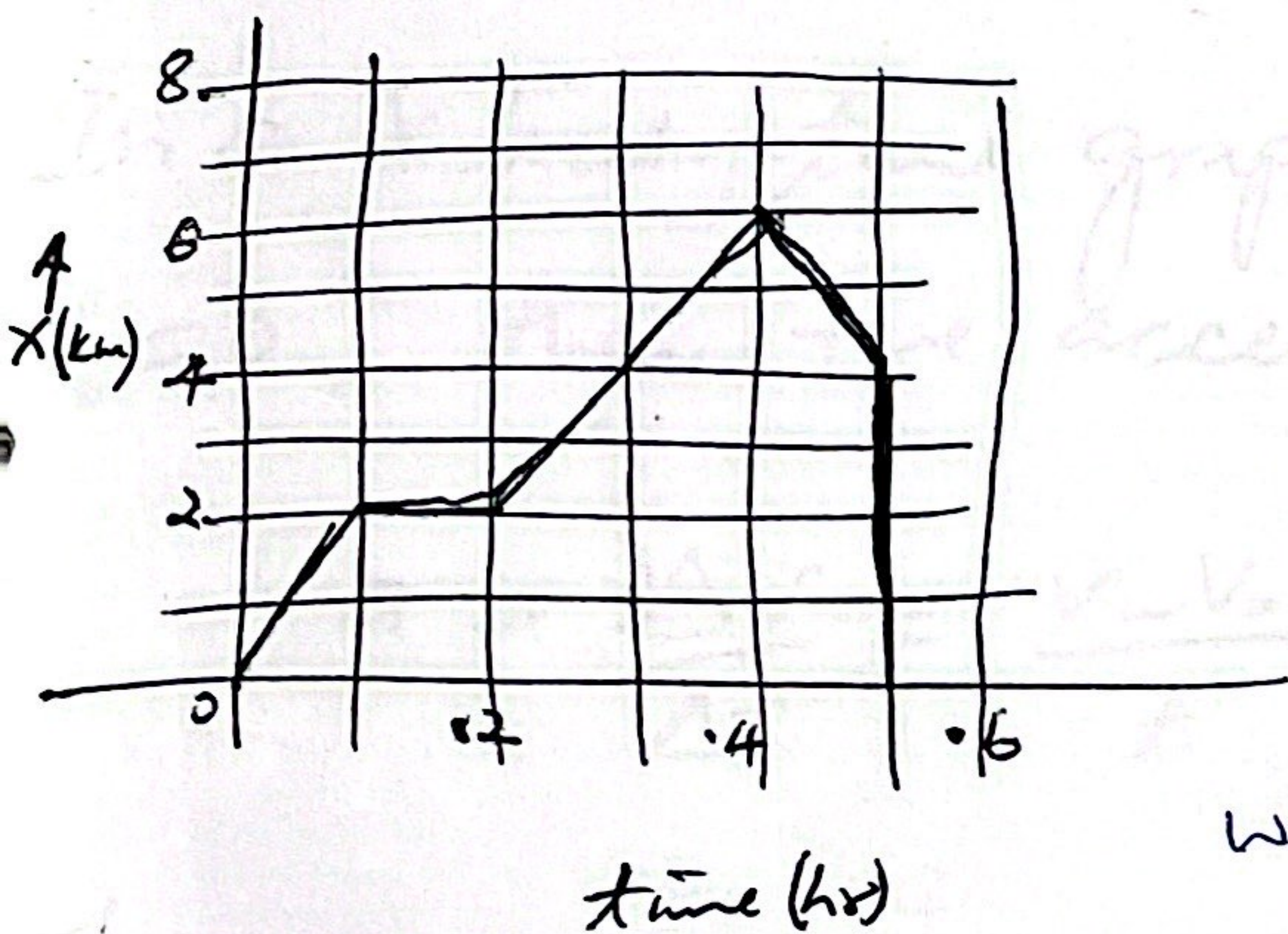
$$\bar{v} = \underline{\underline{30 \text{ mi/hr.}}}$$

5" $\frac{15}{20}$
 $\frac{15}{20}$

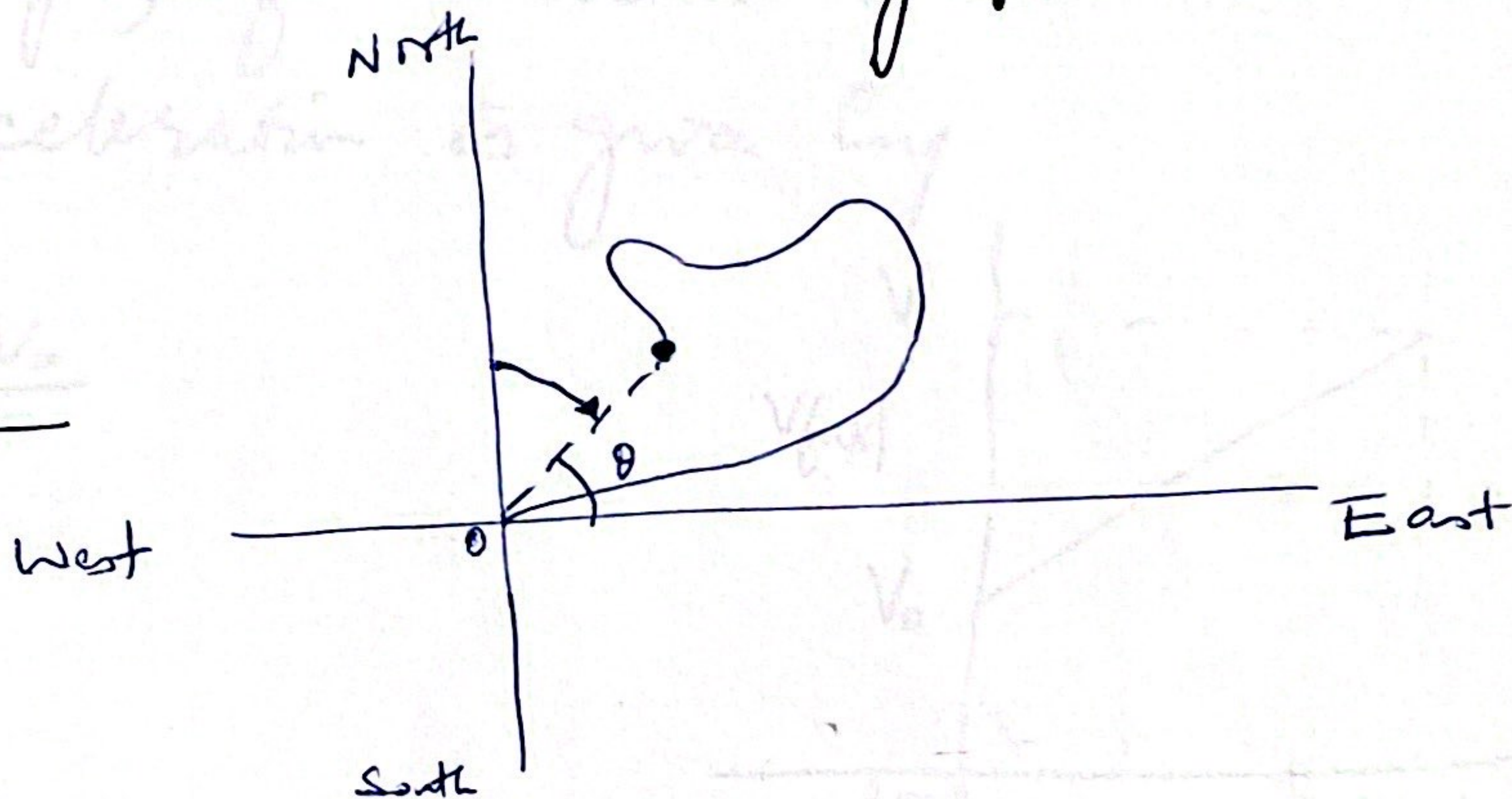
$\frac{15}{20}$

Distance and displacement — Acceleration.

Distance — Space between two points/planes along a path (scalar).
 Displacement — space between two points; origin and terminal (vector).



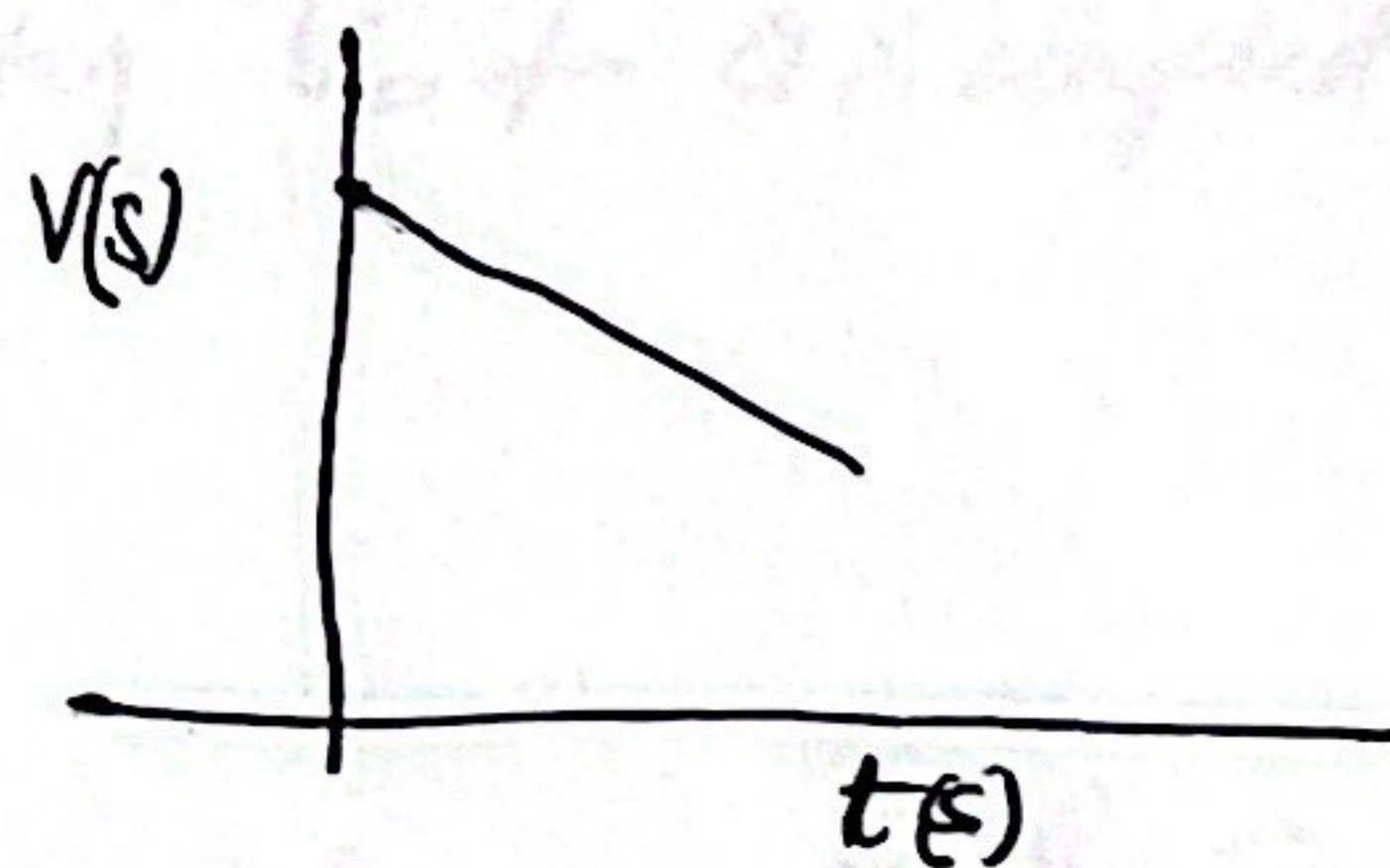
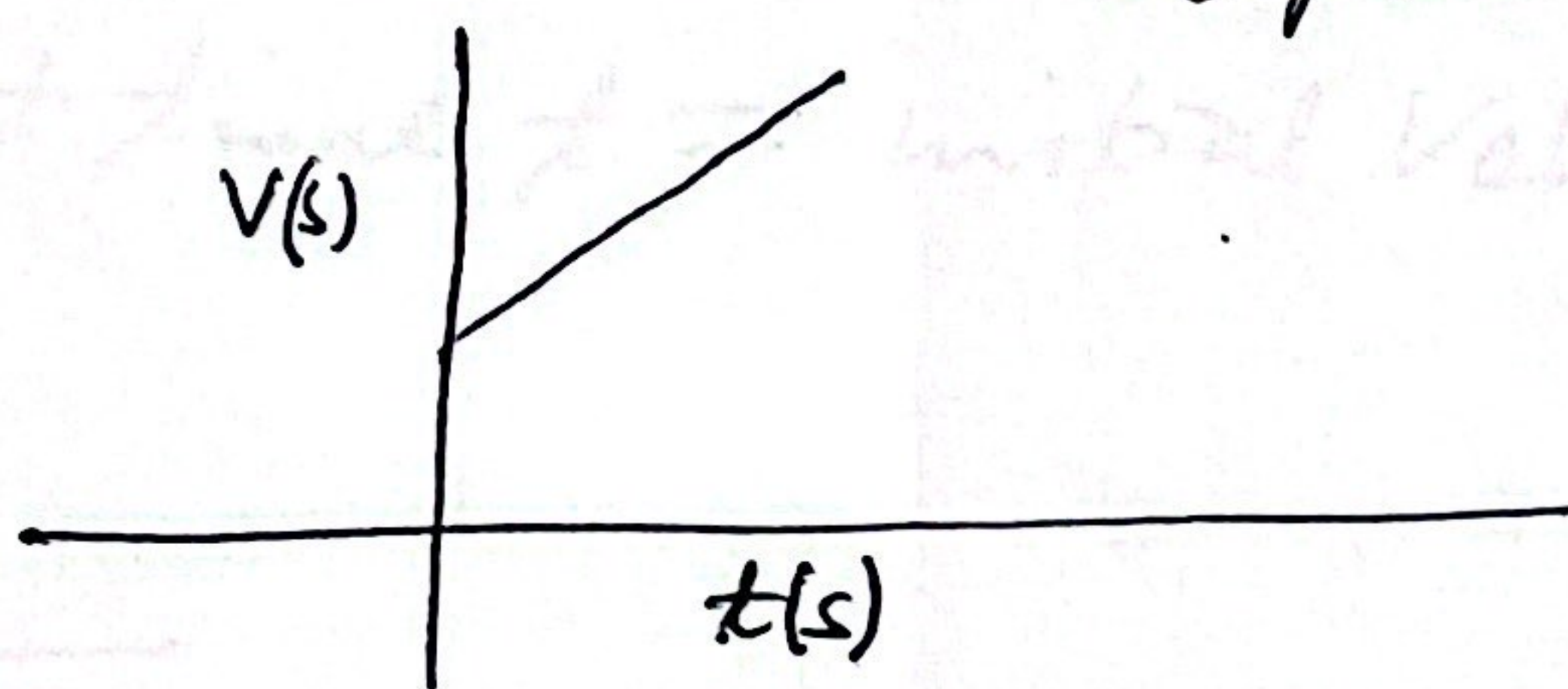
Distance-time graph.



Acceleration is the rate of change of Velocity with time

$$\bar{a} \text{ average acceleration} = \frac{\text{Change in speed}}{\text{elapsed time}} = \frac{V - V_0}{t - t_0}$$

Units m/s^2

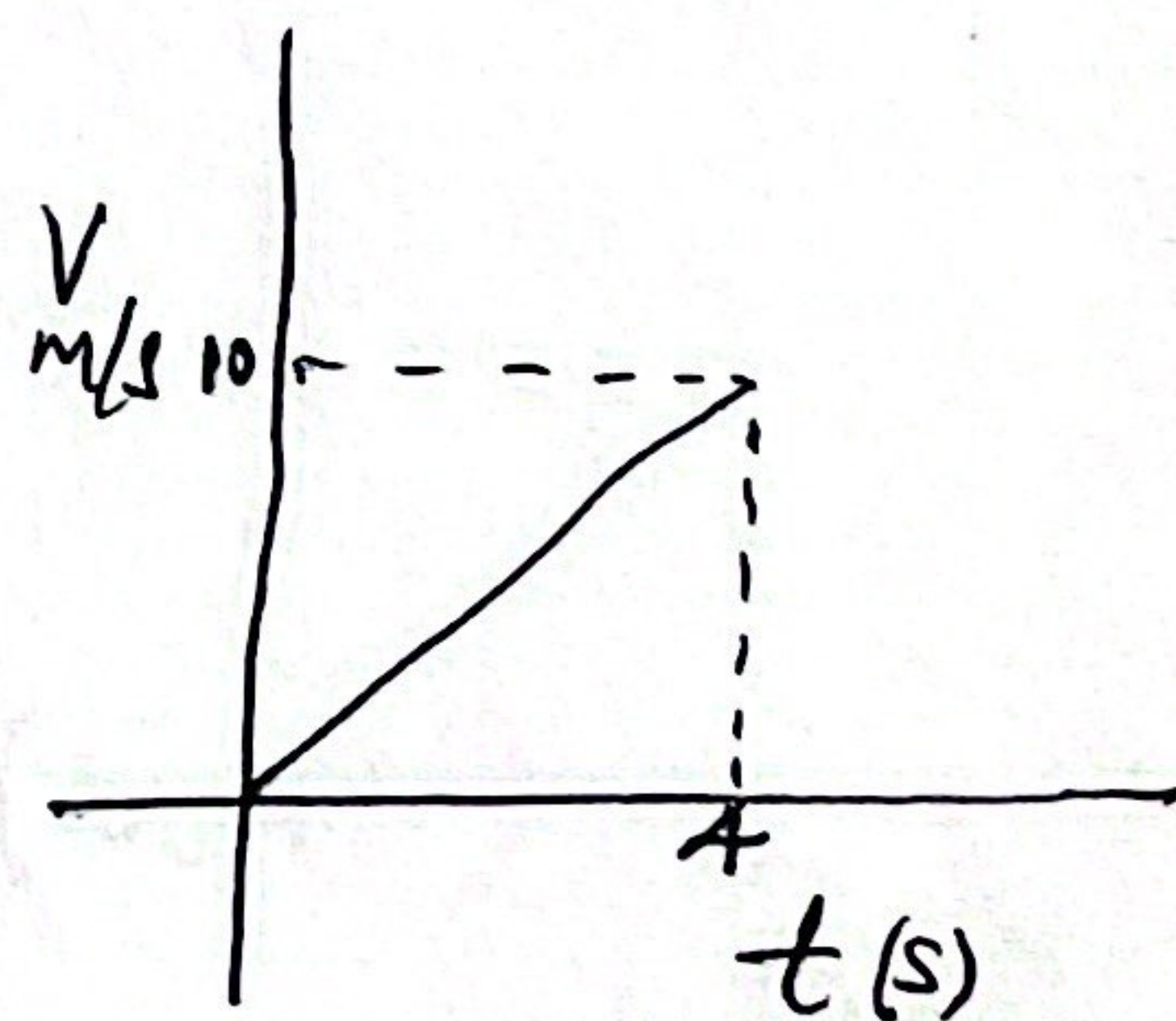


Example.

A man accelerates from rest to 10 m/s in 4 seconds. Find the average acceleration.

Sol.

$$\bar{a} = \frac{10 \text{ m/s} - 0 \text{ m/s}}{4 \text{ s} - 0 \text{ s}} = \frac{10}{4} \text{ m/s}^2 = \underline{\underline{2.5 \text{ m/s}^2}}$$



Newton's ^{equations} ~~laws~~ of motion. 3.2

The simplest type of acceleration found in nature is the acceleration due to gravity g . Near the earth's surface this acceleration is constant in time, $g \approx 9.8 \text{ m/s}^2$.

In a Velocity-time graph, if the initial time occurs at $t_0 = 0$ then the acceleration is given by

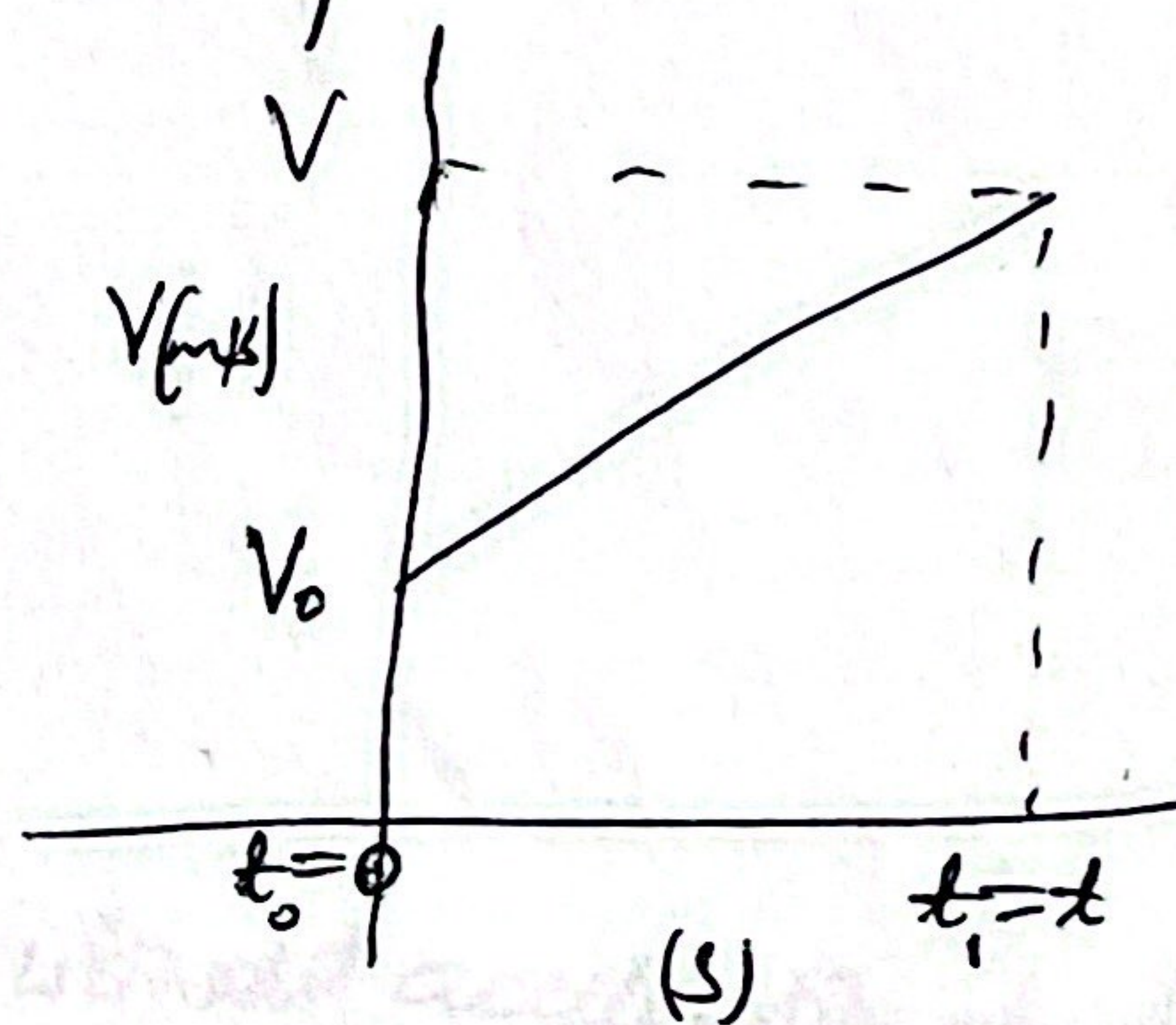
$$a = \frac{\Delta V}{\Delta t} = \frac{V - V_0}{t}$$

from which

$$V - V_0 = at$$

or

$$\boxed{V = V_0 + at}$$



i.e. the velocity at time $t = \text{initial velocity } V_0 + a \times \text{elapsed time}$

Distance travelled

In a constant acceleration a , the distance travelled is given by the product of velocity average and the elapsed time

$$X = \bar{v} t$$

$$\text{Average velocity } \bar{v} = \frac{\text{Initial } V_0 + \text{final } V}{2} = \frac{1}{2} (V_0 + V)$$

$$X = \frac{1}{2}(v_0 + v)t = \frac{1}{2}v_0 t + \frac{vt}{2}$$

$$\begin{aligned} X &= \frac{1}{2}v_0 t + \frac{1}{2}vt \quad v = v_0 + at \\ &= \frac{1}{2}v_0 t + \frac{1}{2}(v_0 + at)t \\ &= \frac{1}{2}v_0 t + \frac{1}{2}v_0 t + \frac{1}{2}at^2 \\ &= \underline{\underline{v_0 t + \frac{1}{2}at^2}} \end{aligned}$$

Since $v = v_0 + at$, then

$$X = \frac{1}{2}v_0 t + \frac{v_0 t}{2} + \frac{1}{2}at^2$$

Or finally

$$\boxed{X = v_0 t + \frac{1}{2}at^2}$$

i.e. distance travelled = distance travelled without acceleration + distance travelled with acceleration.

For the case of uniform acceleration for initial conditions $X=0, v=v_0$ at $t=0$ we can summarize.

Acceleration	:	$a = \text{const} = \frac{v-v_0}{t}$
Velocity	:	$v = v_0 + at$
Distance	:	$X = v_0 t + \frac{1}{2}at^2$

1

2

3

Example

An automobile travelling at a speed of 30 mi/hr accelerates uniformly to a speed of 60 mi/hr in 10 s. How far does the automobile travel during the time

$$u = 30 \text{ m/s}$$

$$t = 10.$$

$$v = 60 \text{ m/s}$$

$$v = u + at$$

$$60 = 30 + a \times 10$$

$$10a = 30$$

$$a = 3 \text{ m/s}^2$$

$$s = ut + \frac{1}{2}at^2$$

$$= 30 \times 10 + \frac{1}{2} \times 3 \times 10^2$$

$$s = 300 + 150 = 450 \text{ m}$$

$$s = ut + \frac{1}{2}at^2$$

$$= 30 \times 10 + \frac{1}{2} \times 3 \times (10)^2$$

$$= 300 + 150 = 450 \text{ m/s}$$

$$X = v_0 t + \frac{1}{2}at^2$$

$$X = v_0 t + \frac{1}{2}at^2$$

Sol.

$$V = V_0 + at$$

Final $V = 60 \text{ mi/hr}$, initial $V = V_0 = 30 \text{ mi/hr}$ time $t = 10 \text{ s}$.

$$60 \text{ mi/hr} = 30 \text{ mi/hr} + a \cdot 10 \text{ s}$$

$$a = \frac{60 \text{ mi/hr} - 30 \text{ mi/hr}}{10 \text{ s}} = \frac{30 \text{ mi/hr}}{10 \text{ s}}$$

$$\left[\begin{array}{l} 30 \text{ mi/hr} = 44 \text{ ft/s} \\ 15 \text{ mi/hr} = 22 \text{ ft/s} \end{array} \right]$$

$$a = \frac{44 \text{ ft/s}}{10} = 4.4 \text{ ft/s}.$$

$$X = V_0 t + \frac{1}{2} at^2$$

$$= 44 \text{ ft/s} \times 10 \text{ s} + \frac{1}{2} \times 4.4 \text{ ft/s}^2 \times (10 \text{ s})^2$$

$$X = 440 \text{ ft} + 220 \text{ ft}$$

$$X = 660 \text{ ft}$$

Exercise

A single air molecule of typical speed 400 m/s is put into a vacuum chamber of $\Phi 2 \text{ m}$. How many times per second will the molecule cross the space between the walls? How far will the molecule go per day?

If time is not known

$$V = V_0 + at$$

$$\text{and } X = V_0 t + \frac{1}{2} at^2$$

$$t = \frac{V - V_0}{a}$$

$$\Rightarrow X = V_0 \left(\frac{V - V_0}{a} \right) + \frac{1}{2} a \left(\frac{V - V_0}{a} \right)^2$$

$$= \frac{V_0 V}{a} - \frac{V_0^2}{a} + \frac{V^2}{2a} + \frac{V_0^2}{2a} - \frac{V V_0}{a}$$

$$X = \frac{V^2}{2a} - \frac{V_0^2}{2a}$$

$$\Rightarrow \boxed{V^2 - V_0^2 = 2aX} \quad \dots \dots \dots 4$$

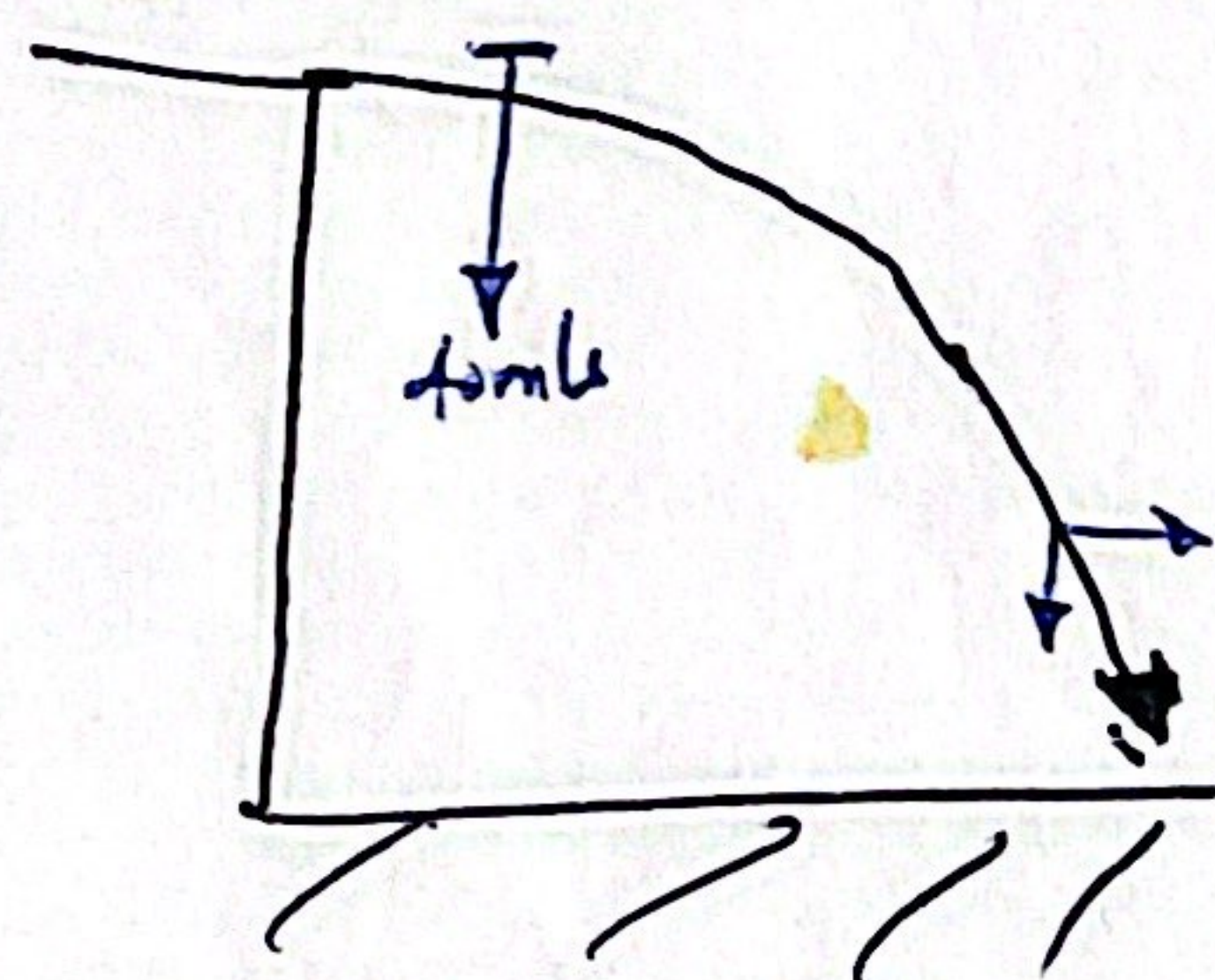
Motion of Falling Bodies

Galileo trying to tackle the problem of falling bodies found that two or more objects fall at almost the same rate. Present day techniques permit verification of Galileo's hypothesis to be made with high precision. From measurements made with such techniques, we find that the acceleration experienced by a body falling freely near the surface

Example.

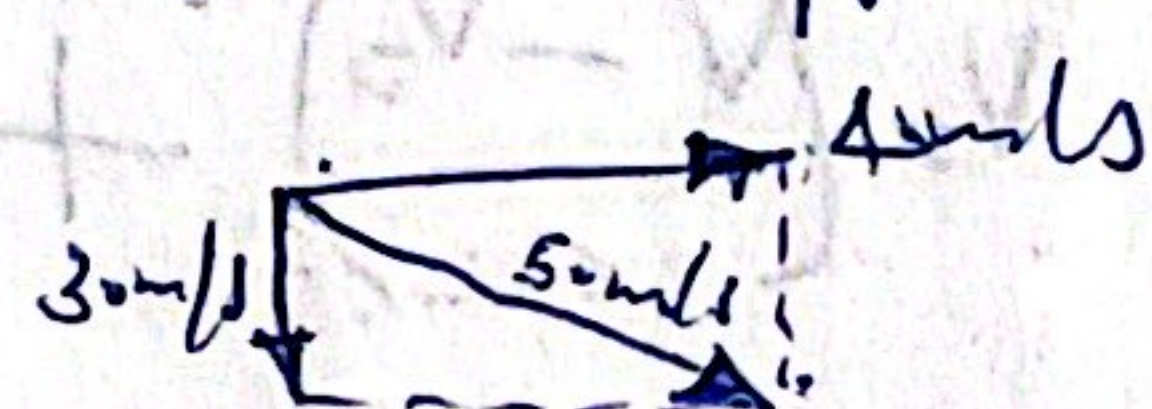
A bullet is fired from the top of a building 45m high with initial speed 40 m/s. How long does it take to reach the ground and what is its speed on impact?

Sol.



$$g = 10 \text{ m/s}^2$$

* ~~40m/s~~ If fired horizontally then the approach is different



$$V = at \quad [a \approx 10 \text{ m/s}^2]$$

$$\text{and } x = \frac{1}{2}at^2$$

$$\Rightarrow t = \sqrt{\frac{2x}{a}} = 3 \text{ s}$$

$$V = a \cdot \sqrt{\frac{2x}{a}} = 30 \text{ m/s}$$

$$\text{speed on impact} = \sqrt{40^2 + 30^2}$$

$$= \sqrt{2500} = 50 \text{ m/s}$$

$$V^2 - u^2 = 2ah \text{ gives final speed}$$

$$\Rightarrow V = \sqrt{2ah + u^2} = \sqrt{40^2 + 2 \times 10 \times 45} = 50 \text{ m/s}$$

The time required is ~~given~~ from $V = u + at$

$$\Rightarrow t = \frac{V - u}{a} = \frac{(50 - 40) \text{ m/s}}{10 \text{ m/s}^2} = 1 \text{ s}$$

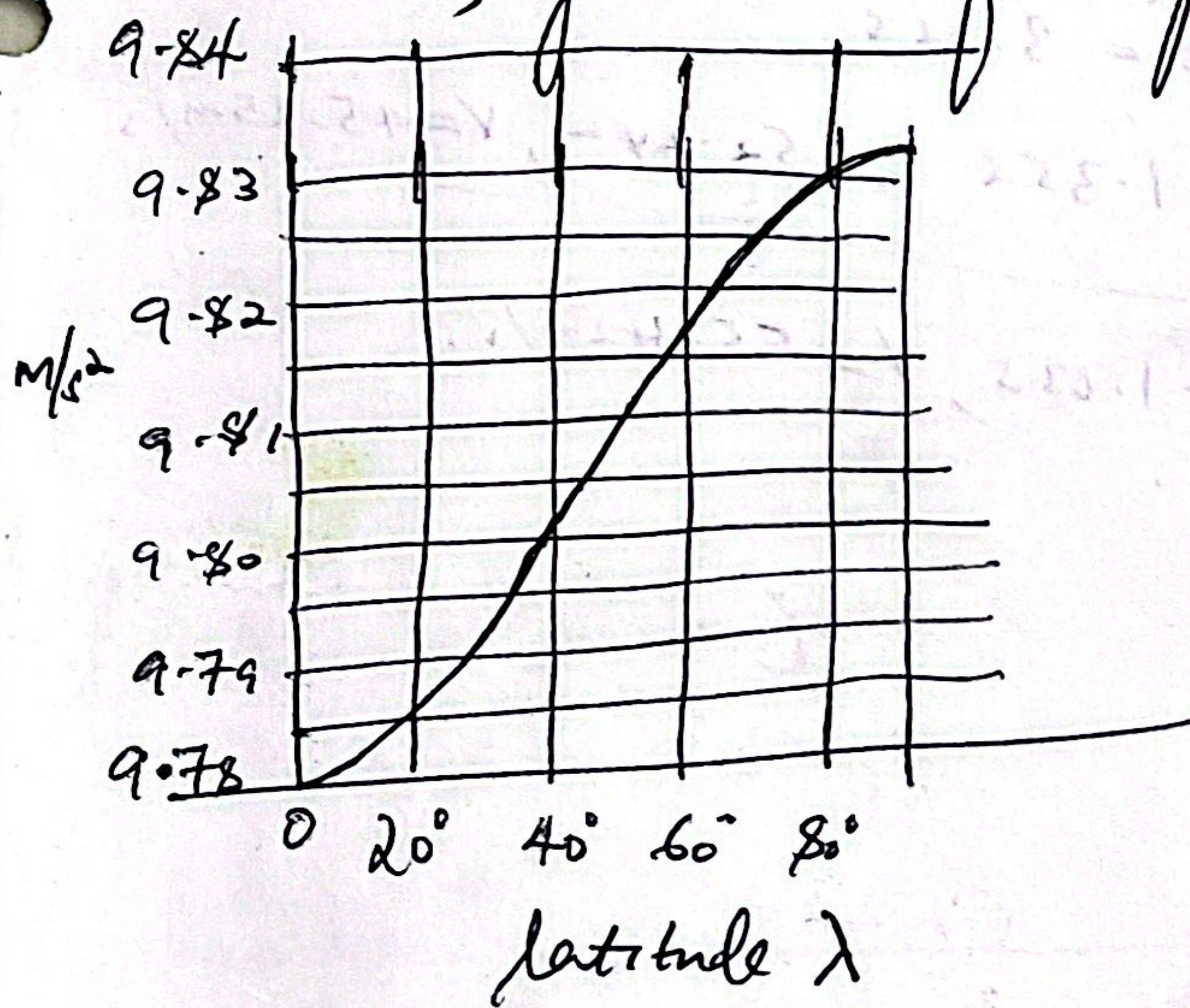
Exercise

A racing car leaving at 90 m/s has 100m to go in the race and must take the distance in 30s to break record for the course. What constant acceleration is required and what final speed does the car have?

of the Earth is approximately 9.8 m/s^2 or 32 ft/s^2 .

$$g = 9.8 \text{ m/s}^2 = 32 \text{ ft/s}^2$$

Because Earth is not a homogeneous sphere (and also because it rotates), g varies from place to place on the surface.



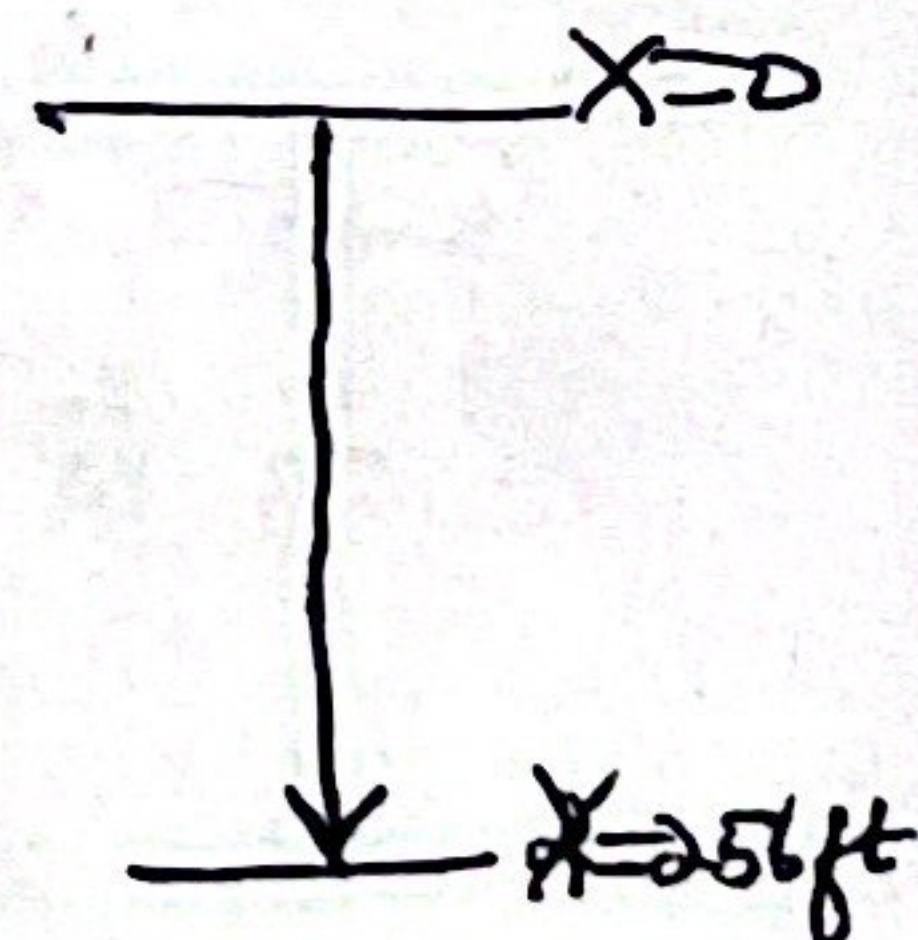
Example

A ball is released from rest at a certain height. What is the velocity after falling 256 feet?

Sol.

Since $v_0 = 0$ and $a = g$ then $x = 0 \cdot t + \frac{1}{2}gt^2$

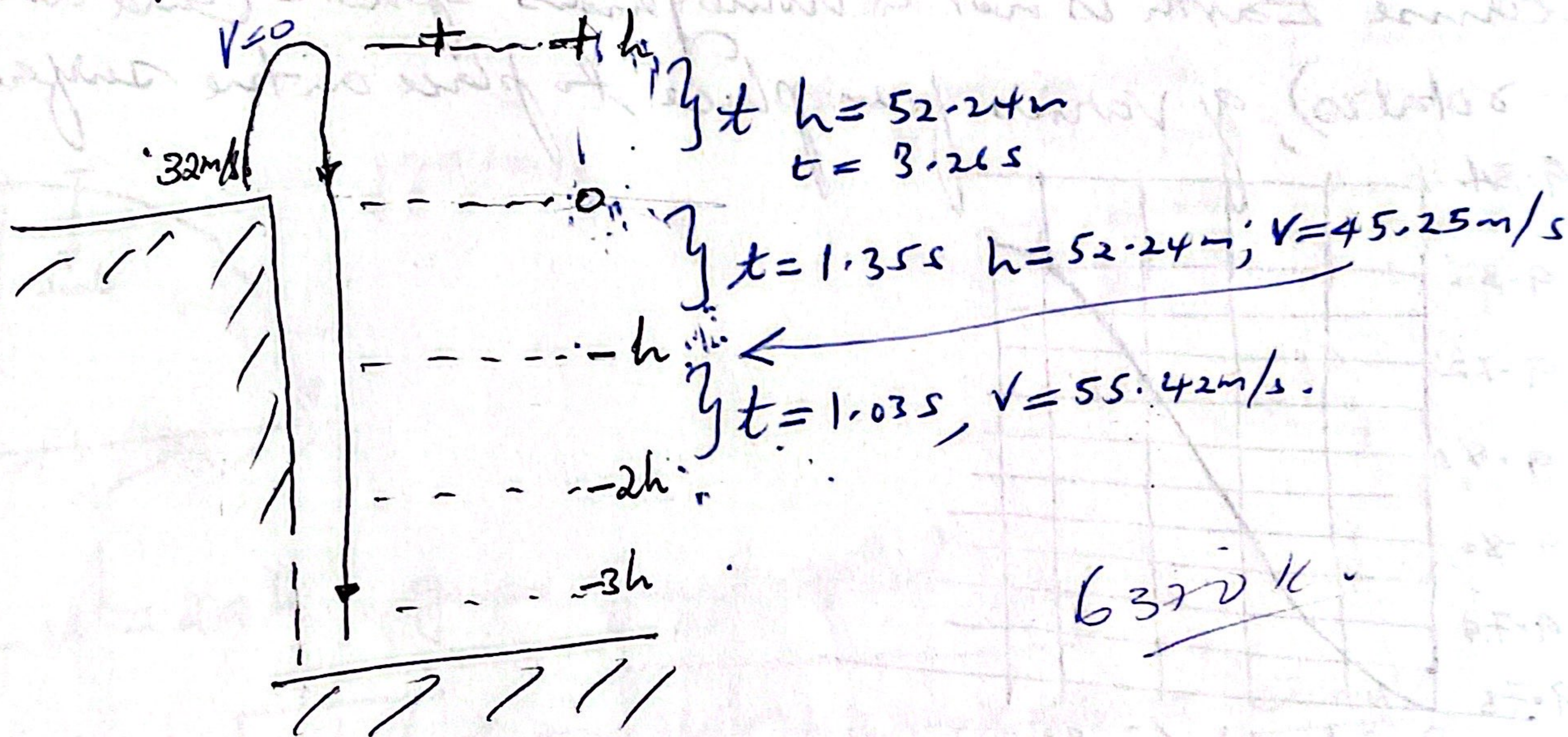
Solving for $t = \sqrt{\frac{2x}{g}}$. The velocity for falling for a time t is $v = gt = g \times \sqrt{\frac{2x}{g}} = \sqrt{2gx} = \underline{\underline{128 \text{ ft/s}}}$



Exercise.

A ball is thrown upward with an initial velocity of 32 m/s from the top of a building. Calculate the velocity and position as functions of the time.

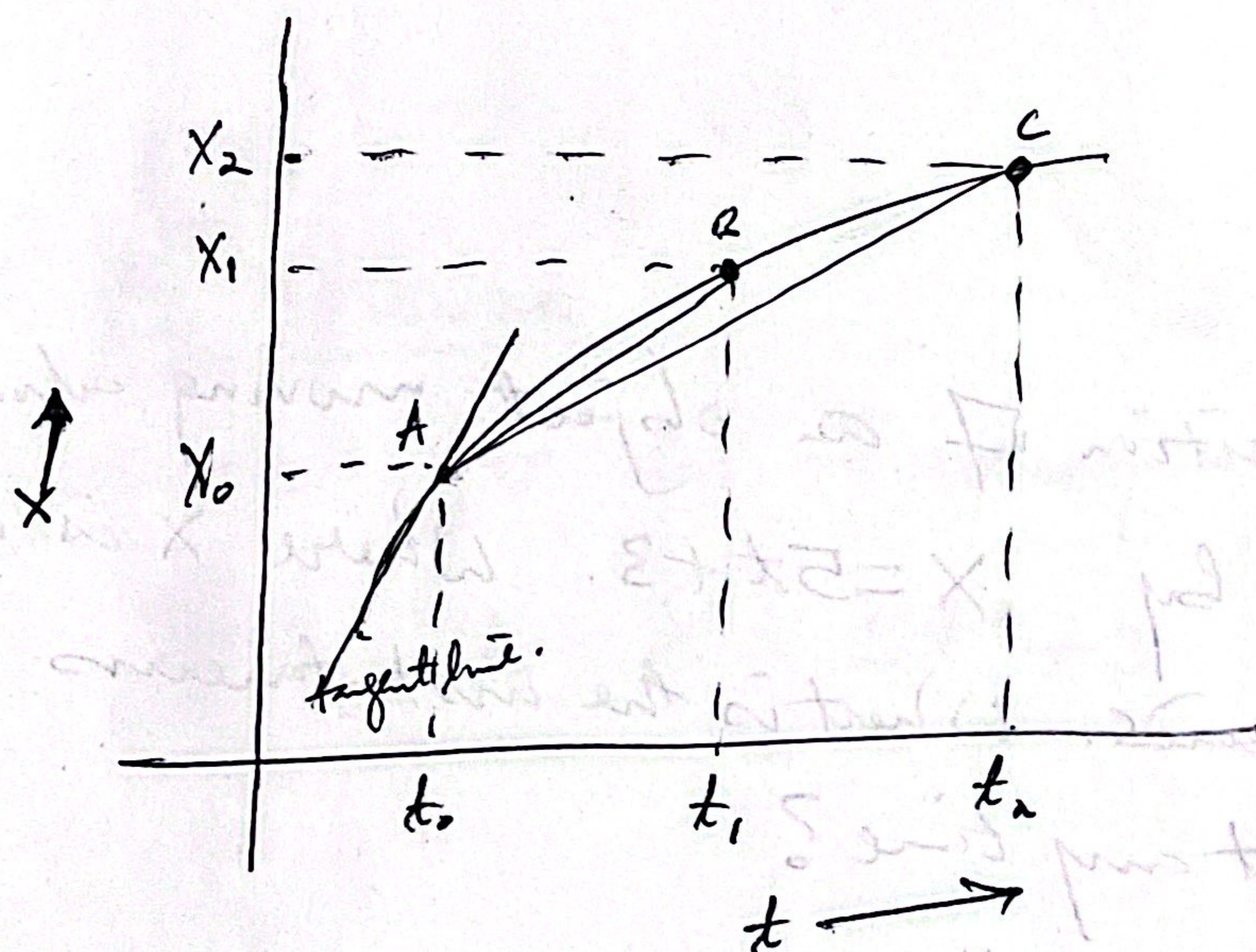
Hint: $h = \text{height}$.



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Instantaneous Velocity and acceleration

If the motion of an object does not take place with constant velocity, then in general, the average velocity depends on the time interval chosen for calculation. The instantaneous velocity results when the ~~large~~ time interval is infinitesimally small and is equal to the slope of the tangent line at a point.



Distance-time graph.

Average Velocity between A and C is $\bar{V}_{02} = \frac{x_2 - x_0}{t_2 - t_0}$

A and B $\bar{V}_{01} = \frac{x_1 - x_0}{t_1 - t_0}$

By observation $\bar{V}_{01} > \bar{V}_{02}$

If we take other intervals small of time eventually we arrive at the smallest interval possible that will not alter the average velocity. This limiting average velocity is the

instantaneous Velocity, v . It is expressed as

$$V = \lim_{t_1 \rightarrow t_0} \frac{x_1 - x_0}{t_1 - t_0} = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t}.$$

This equation states that the instantaneous velocity v is given by the ratio of Δx to Δt in the limit that Δt approaches zero.

Example

Suppose that the position of an object moving along the x -axis is given by $x = 5t + 3$. Where x is in metres when t is in seconds. What is the instantaneous velocity of the object at any time?

Sol.

At time t_0 , the position is $x_0 = 5t_0 + 3$

and at time $t_1 = t_0 + \Delta t$, the position is $x_1 = 5t_1 + 3$

$$x_1 = 5(t_0 + \Delta t) + 3$$

Therefore

$$V = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{[5(t_0 + \Delta t) + 3] - [5t_0 + 3]}{\Delta t}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{(5t_0 + 5\Delta t + 3) - (5t_0 + 3)}{\Delta t}$$

$$V = \lim_{\Delta t \rightarrow 0} \frac{5\Delta t}{\Delta t}$$

$$\underline{\underline{V = 5 \text{ m/s.}}}$$

In this case, Δt cancels in the expression for V ; that is the instantaneous velocity is not a function of time and is constant (and is equal to the average velocity).

(Q5)

Exercise.

* Repeat for an object moving and the position is given by $X = 5t^2 + 3$